

Name: Solar Panel Team

Date: 10/10/2025

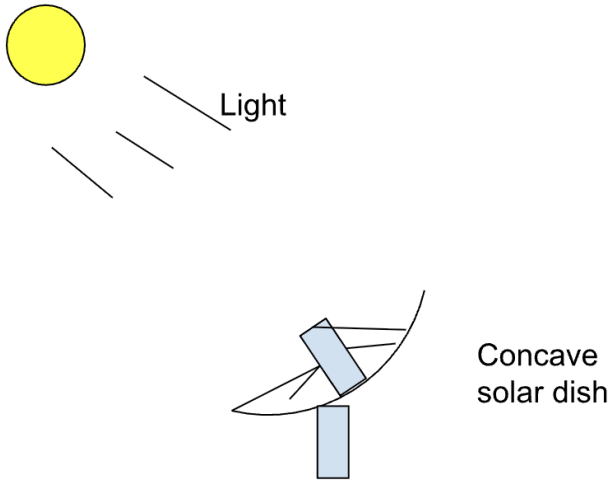
Class: Design for Engineers e4usa

MyDesign Portfolio

Element D Initial Template AT - 15 points

Element D: Design Concept Generation, Analysis, and Selection

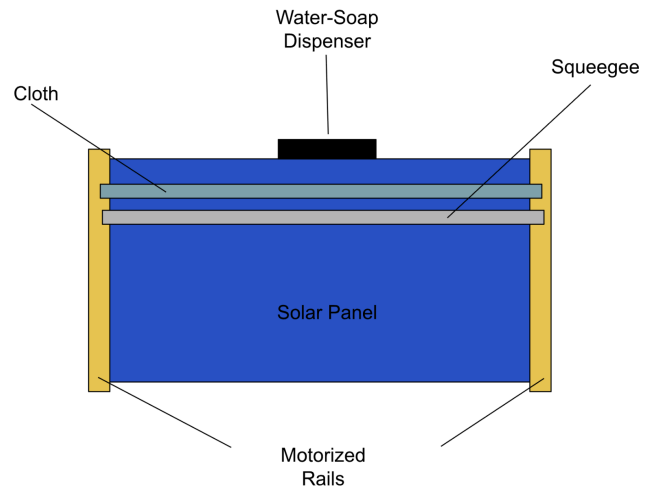
Design Idea Generation (D1)

Design Idea #1	New Solar Panel Geometry
Date:	10/16/25
Description:	The design concept comes from radio astronomy, where solar panels are reshaped to capture more light than flat rectangular ones.
Drawing that includes distinguishing information such as scale, dimensions, and materials:	

Name: Solar Panel Team

Date: 10/10/2025

Class: Design for Engineers e4usa

Design Idea #2	Solar Panel squeegee
Date:	10/16/25
Description:	This design concept comes from car windshield wiper mechanisms installed on panels capable of automated cleaning.
Drawing that includes distinguishing information such as scale, dimensions, and materials:	

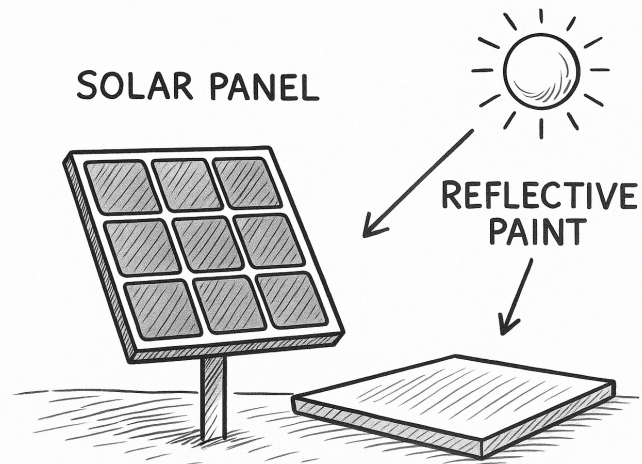
Design Idea #3	Reflective paint
Date:	10/22/25
Description:	This design concept relies on reflective paint on surfaces around the solar panel to increase sun reflection on to increase efficiency.

Name: Solar Panel Team

Date: 10/10/2025

Class: Design for Engineers e4usa

Drawing that includes distinguishing information such as scale, dimensions, and materials:

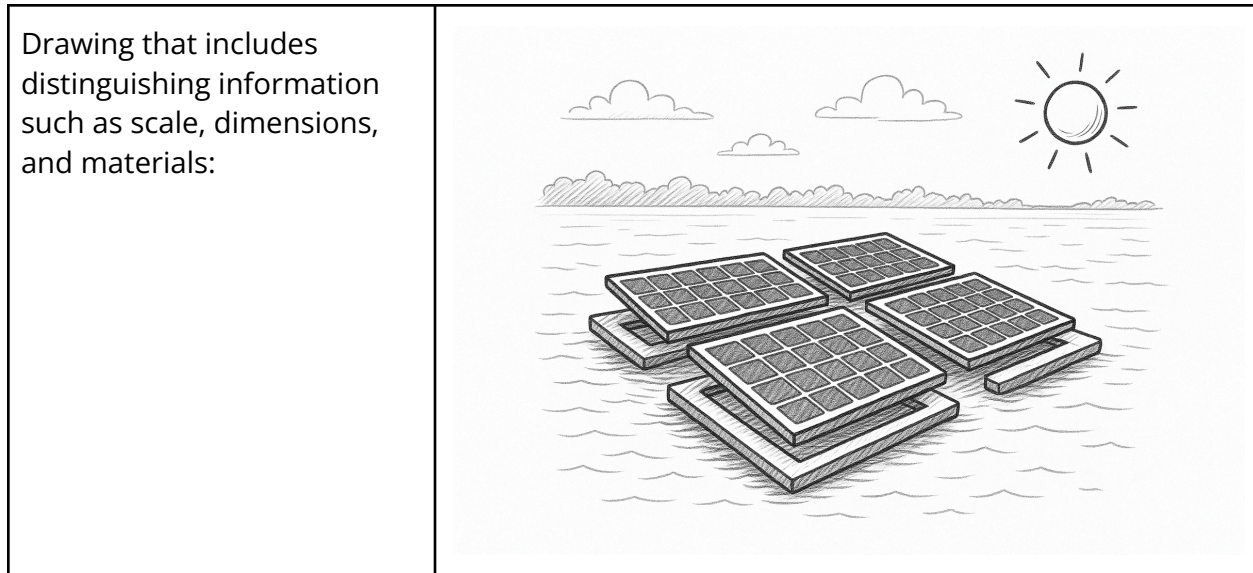


Design Idea #4	Floating solar panels
Date:	10/22/25
Description:	Solar panels that could float on water can boost efficiency by having more stable temperatures.

Name: Solar Panel Team

Date: 10/10/2025

Class: Design for Engineers e4usa



Design Idea Comparison and Selection (D1, D2)

(List your requirements from Element A below)	Weight (multiplied by each design score)	New Solar Panel Geometry	Solar Panel Squeegee	Reflective Paint	Floating Solar Panels
Requirement A: This device should be able to operate and swipe over a flat surface.	10	0	10	10	10
Requirement B: The design should be stable and hold itself securely to the panel.	15	10	15	15	5
Requirement C: The design should be able to withstand rough environmental	15	15	15	10	0

Name: Solar Panel Team

Date: 10/10/2025

Class: Design for Engineers e4usa

conditions (blizzards, sandstorms, dirt, etc) that can cause weather to affect a solar panel.					
Requirement D: This device should be able to excrete water and cleaning solutions.	10	0	10	0	0
Requirement E: This device should be able to operate (come into contact with the panel) without damaging the panel itself.	10	5	10	10	5
Requirement F: The design should improve the solar panel efficiency.	25	15	25	15	5
Requirement G: The design should be creative and aesthetic.	5	5	5	5	5
Total score	90	50	90	65	30

Our team chose the Solar Panel Squeegee (design #2) to solve the community problem because the design incorporates all design requirements, such as improving solar efficiency, operating properly on the panel with low maintenance, and reaching our design goals.

Design Blueprint (D4)

Design Blueprint #1

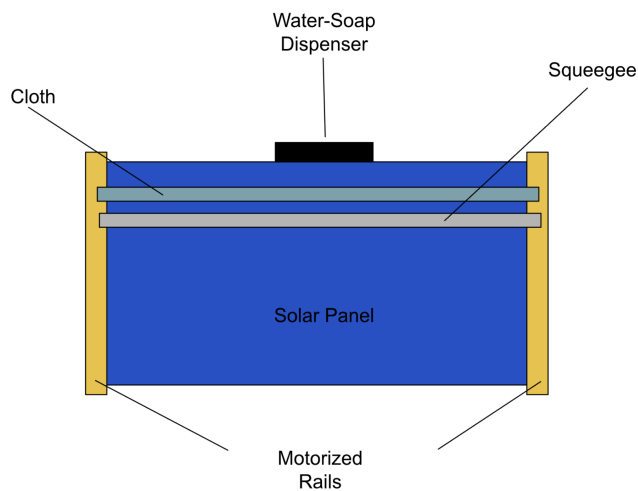
Name: Solar Panel Team

Date: 10/10/2025

Class: Design for Engineers e4usa

Date: 10/23/25

Sketch or CAD Drawing (orthographic):



Manufacturing Methods (cut, saw, paste, screw, tie, glue):

Step 1: Install Rails

Step 2: Manufacture the device

Step 3: Add wiper/sprinkler

Supply List and Costs:

Item	Cost each	# needed	Total
Guiding rails	\$4	2	8\$
Squeegee	\$2	1	\$2
Cloth	\$1	1	\$1
Total Cost	\$7	4	\$11

Name: Solar Panel Team

Date: 10/10/2025

Class: Design for Engineers e4usa

Plan of Action (D3)

	TASK TITLE	TASK OWNER - Who is in charge of the task	START DATE	DUE DATE	DURATION - How long should it take to complete	PERCENT OF TASK COMPLETE	
1	Project Conception and Initiation						M
3.2.3	Team Charter	Whole team			1 day	100%	
3.3.3	Team Charter Revisions	whole team			1 day	100%	
3.5.1	Research - for Element A and B	whole team			1 day	100%	
3.4.1	Stakeholders - Who communicates with them?	W			1 day	100%	
	Element A	B			.5 day	100%	
	Element B	S			.5 day	100%	
	Element C	A			.5 day	100%	
2	Project Planning						
3.6	Brainstorming and Selecting Ideas	whole team			1 day	100%	
3.7.1	Engineering Drawings	whole team			1 day	100%	
3.8.3	Communication Plan - Google, Group chat, Notebook	whole team			1 day	100%	
	Risk Management - What could go wrong with the Prototype	whole team			1 day	100%	
	Element D	whole team			1 day	100%	
	Element E	whole team			1 day	100%	
	Element F	whole team			1 day	100%	

The Team Ghant chart is placed above to show our design planning.

Iterations for the Solution (How did you change your prototype after building and testing it?):

Date of Iteration: 1/06/26

Description of Iteration: Changed the dimensions and geometry of the design to accommodate a proper fit.

Images if applicable:

